

LOUIS TIAO

RESEARCH SCIENTIST, META

[GitHub](#) · [Google Scholar](#) · [LinkedIn](#)

[tiao.io](#) · louis@tiao.io

Research scientist working on Bayesian optimization, Gaussian processes, and sample-efficient methods for automated machine learning. PhD from the University of Sydney, with work recognized at NeurIPS and ICML through Oral and Spotlight presentations. Several years of industrial research across Meta, Amazon, and Secondmind, building open-source software used at scale.

EXPERIENCE

- Aug 2024 – Present* **Research Scientist**, Meta · New York, US
Central Applied Science (CAS), Adaptive Experimentation team. Develop sample-efficient Bayesian optimization and AutoML methods (multi-fidelity optimization and scaling-law modeling) within the open-source Ax and BoTorch frameworks, applied to hyperparameter tuning and capacity management for large-scale Ads ranking models. Co-first author on *Ax: A Platform for Adaptive Experimentation* (AutoML 2025).
- May 2022 – Sep 2022* **Applied Scientist Intern**, Amazon Web Services · Cambridge, UK
Led an exploratory research project on hyperparameter optimization for large language models, studying LLM scaling behavior and whether optimal hyperparameters extrapolate from small models to large ones. Hosted by Aaron Klein, Matthias Seeger, and Cédric Archambeau.
- Sep 2021 – Apr 2022* **Doctoral Student Researcher**, Secondmind · Cambridge, UK
An AI research lab (formerly Prowler.io) focused on Bayesian optimization and Gaussian processes. Contributed open-source software for efficient GP sampling and led research on integrating neural-network features into GP approximations, which led to *Spherical Inducing Features for Orthogonally-Decoupled Gaussian Processes* (ICML 2023, Oral). Worked closely with Vincent Dutoit and Victor Picheny.
- Jun 2019 – Dec 2019* **Applied Scientist Intern**, Amazon Web Services · Berlin, Germany
Contributed to Automatic Model Tuning in Amazon SageMaker, focusing on Bayesian optimization for AutoML. Led research integrating multi-fidelity BO with asynchronous parallelism, resulting in a paper and open-source code in AutoGluon that later formed the basis of SyneTune. Hosted by Matthias Seeger, Cédric Archambeau, and Aaron Klein.
- Jun 2016 – Apr 2019* **Research Software Engineer**, CSIRO's Data61 · Sydney, Australia
On the Inference Systems Engineering team at Australia's national AI research division, built microservices and open-source libraries for large-scale Bayesian deep learning. A stint with the Graph Analytics team produced research on graph representation learning and a NeurIPS spotlight.
- Jun 2015 – Jun 2016* **Research Software Engineer**, NICTA · Sydney, Australia
Worked within an interdisciplinary ML research team on the Big Data Knowledge Discovery initiative, developing and releasing open-source libraries for applying Bayesian ML at scale. NICTA later merged into CSIRO's Data61.
- Aug 2017 – Nov 2018* **Teaching Assistant**, UNSW · Sydney, Australia
Taught COMP9418 (Advanced Topics in Statistical Machine Learning), a postgraduate course on probabilistic graphical models and approximate inference.

EDUCATION

- 2024 **Ph.D., Computer Science**, University of Sydney
Thesis: *Probabilistic Machine Learning in the Age of Deep Learning: New Perspectives for Gaussian Processes, Bayesian Optimization and Beyond*. Supervised by Fabio Ramos and Edwin Bonilla.
- 2015 **B.Sc. (Honours, 1st Class), Computer Science**, UNSW Sydney
Concentrations: Artificial Intelligence and Mathematics.

SELECTED PUBLICATIONS

- 2026 J. A. Lin, S. Ament, **L. Tiao**, D. Eriksson, M. Balandat, E. Bakshy. *Empirical Gaussian Processes*. ICML2026.
- 2025 M. Olson, E. Santorella, **L. Tiao**, et al. *Ax: A Platform for Adaptive Experimentation*. AutoML2025, ABCD Track.
- 2023 **L. Tiao**, V. Dutordoir, V. Picheny. *Spherical Inducing Features for Orthogonally-Decoupled Gaussian Processes*. ICML2023. **Oral**
- 2022 R. Oliveira, **L. Tiao**, F. Ramos. *Batch Bayesian Optimisation via Density-Ratio Estimation with Guarantees*. NeurIPS2022.
- 2021 **L. Tiao**, A. Klein, C. Archambeau, E. V. Bonilla, M. Seeger, F. Ramos. *Bayesian Optimization by Density-Ratio Estimation*. ICML2021. **Oral**
- 2020 P. Elinas, E. V. Bonilla, **L. Tiao**. *Variational Inference for Graph Convolutional Networks in the Absence of Graph Data and Adversarial Settings*. NeurIPS2020. **Spotlight**
- Full list at tiao.io/publications and Google Scholar.

SELECTED OPEN-SOURCE SOFTWARE

- 2024– **Ax / BoTorch**: platforms for adaptive experimentation and Bayesian optimization in PyTorch (Meta).
- 2021– **GPflux**: a TensorFlow/Keras framework for Deep Gaussian Processes (Secondmind).
- 2021– **BORE**: Bayesian optimization by probabilistic classification.
- 2019– **AutoGluon**: an AutoML toolkit for deep learning; core developer of the GP-based multi-fidelity searcher (AWS).

TECHNICAL SKILLS

Languages: Python (expert), C, Java, Bash, R, MATLAB, Haskell.

Machine learning: PyTorch, TensorFlow / TF Probability, JAX, scikit-learn, GPyTorch, GPflow, Hugging Face.

Scientific computing: NumPy, SciPy, pandas, Matplotlib, Jupyter.

Tools & infrastructure: Git, Docker, Kubernetes, Hydra, AWS, Slurm / HPC, $\LaTeX$.

ACADEMIC SERVICE

- 2021– **Conference reviewing:** NeurIPS, ICML, ICLR, AISTATS, UAI, and AutoML-Conf, plus workshops at NeurIPS, ICML, and ICLR. **Best Reviewer Award** (top 10%), ICML 2021.
- 2019– **Journal reviewing:** Transactions on Machine Learning Research (TMLR) and IEEE TPAMI.